SPINE · NEET-SS / FET / INI-SS

Cervical trauma

1. Clearance and imaging

- **NEXUS criteria** (no imaging needed if all five absent): **midline cervical tenderness, focal neurological deficit, altered alertness, intoxication, painful distracting injury**.
- **Canadian C-spine rule:** high-risk factors (**age 65 or more, dangerous mechanism, paraesthesia in the limbs**) = image; low-risk factors allowing range-of-motion assessment; able to rotate the neck 45 degrees left and right = no imaging. More sensitive and specific than NEXUS.
- **CT is the first-line imaging** in adult trauma (radiographs miss injuries, especially at the cervicothoracic junction and craniocervical junction). **MRI** for neurological deficit, suspected ligament/disc injury (**before reducing a facet dislocation in an obtunded patient**), SCIWORA, and for obtunded patients (debated: CT alone is sufficient to clear many obtunded patients with a high-quality negative CT).
- **Normal values: atlantodens interval (ADI) up to 3 mm in adults (5 mm in children)**; prevertebral soft tissue **under 6-7 mm at C2** and **under 22 mm at C6**; **Powers ratio** (BC/OA) under 1 (over 1 = anterior atlanto-occipital dislocation); **basion-dens interval and basion-axial interval under 12 mm (Harris "rule of 12")**.

2. Craniocervical junction

- **Occipital condyle fractures (Anderson-Montesano):** I comminuted (impaction), II extension of a basilar skull fracture, **III avulsion by the alar ligament** (potentially unstable). Most: collar; occipitocervical fusion if unstable.
- **Atlanto-occipital dissociation:** usually fatal; **children** more often (large head, horizontal condyles). **Traynelis types** (anterior,

longitudinal/distraction, posterior). Detect with **Powers ratio, BDI/BAI**, CT; **never apply traction**; occipitocervical fusion.

3. Atlas (C1) fractures

- **Jefferson (burst) fracture: axial load**, four-part ring fracture (anterior and posterior arches). **Rule of Spence:** combined lateral mass overhang **over 6.9 mm (about 7 mm; 8.1 mm with radiographic magnification)** on the open-mouth view suggests **transverse atlantal ligament (TAL) rupture** (unstable). MRI shows the TAL directly; **Dickman classification of TAL injury** (type I midsubstance rupture: needs **C1-C2 fusion**; type II bony avulsion of the TAL insertion: may heal with immobilisation).
- **Treatment:** stable (intact TAL): **rigid collar/halo**; unstable: **halo** or **C1-C2 (or occipitocervical) fusion**.
- **Neurological injury is rare** (canal widens).

4. Axis (C2) fractures

Odontoid (dens) fractures: Anderson-D'Alonzo (with Grauer modification)

Type	Location	Stability / union	Treatment
I	**Tip avulsion (alar ligament)**, above the transverse ligament	Usually stable (exclude **atlanto-occipital dissociation**)	Collar
II	**Junction of dens and body (waist)**	**Commonest; highest non-union** (watershed blood supply, small contact area)	See below
III	Extends into the **C2 body** (cancellous)	Good union	**Collar or halo**

- **Grauer subtypes of type II: IIA** transverse undisplaced (collar/halo), **IIB** oblique from **anterosuperior to posteroinferior** (or transverse

displaced) = suitable for **anterior odontoid screw** (fracture line perpendicular to the screw), **IIC** oblique from **anteroinferior to posterosuperior** (or comminuted) = **posterior C1-C2 fusion** (screw would shear the fracture).

- **Non-union risk factors for type II: displacement over 5 mm** (also over 4-6 mm), **posterior displacement**, **angulation over 10 degrees**, **age over 50 (about 40-60 in studies)**, smoking, delay in treatment, **comminution**.
- **Elderly (most common cervical fracture over 65):** halo poorly tolerated (aspiration, pneumonia, mortality); options: **rigid collar** (accept stable fibrous union), or **posterior C1-C2 fusion** (Harms: **C1 lateral mass + C2 pars/pedicle screws**; Magerl **C1-C2 transarticular screws**; Gallie/Brooks wiring) for younger elderly patients fit for surgery.
- **Anterior odontoid screw:** needs an **intact TAL**, reducible fracture with a favourable obliquity, and a body habitus allowing the trajectory (**short neck, barrel chest, kyphosis contraindicate it**); osteoporotic bone fails. Preserves C1-C2 rotation (C1-C2 provides about **50% of cervical rotation**).
- **Os odontoideum:** smooth, well-corticated ossicle separated from a hypoplastic dens (probably old unrecognised fracture or developmental); unstable if **sagittal plane translation**: posterior C1-C2 fusion if instability or neurological signs.

Hangman's fracture (traumatic spondylolisthesis of the axis)

- **Bilateral pars (isthmus) fractures of C2**; mechanism **hyperextension with axial load** (diving, car crashes; judicial hanging is hyperextension-distraction). **Neurological injury is uncommon** (canal widens).
- **Levine-Edwards classification:**

Type	Features	Mechanism	Treatment
I	**Minimal translation (under 3 mm; Campbell's: 0-2 mm), no angulation** (vertical fracture line)	Hyperextension + axial load	**Rigid collar**
II	**Over 3 mm translation + significant angulation** (C2-3 disc disrupted)	Hyperextension then flexion	**Reduction (traction in extension) + halo**
IIA	**Severe angulation, minimal translation** (oblique fracture; C2-3 disc and PLL disrupted)	**Flexion-distraction**	**Avoid traction (worsens it)**; reduce with **compression and extension**, halo
III	Pars fracture + **bilateral C2-3 facet dislocation**	Flexion-compression	**Open reduction and fusion** (C2-3 posterior or anterior)

5. Subaxial cervical spine (C3-C7)

Classification

- **SLIC (Subaxial Injury Classification and Severity score, Vaccaro):**

Component	Score
Morphology: no abnormality 0; **compression 1**; burst +1 (=2); **distraction 3**; **rotation/translation 4**	0-4
Disco-ligamentous complex: intact 0; **indeterminate 1**; **disrupted 2**	0-2
Neurology: intact 0; **root injury 1**; **complete cord injury 2**; **incomplete cord injury 3**; **continuous cord compression in the setting of a neurological deficit +1**	0-4

  - **Total under 4: non-operative; 4 = surgeon's discretion; over 4 (5 or more): surgery.**

- **AO Spine subaxial classification:** A (compression), B (tension band injury), C (translation), with facet injury and neurology modifiers.
- **Allen-Ferguson (mechanistic):** compressive flexion, vertical compression, distractive flexion (facet dislocations), compressive extension, distractive extension, lateral flexion.

Specific injuries

- **Facet dislocations (distractive flexion):**
  - **Unilateral facet dislocation:** about **25% anterolisthesis**; rotation; **radiculopathy** commonly; often stable-ish but needs reduction.
  - **Bilateral facet dislocation:** about **50% or more anterolisthesis**; **high rate of cord injury**; very unstable.
  - **Reduction: awake, cooperative patient: closed traction reduction** (Gardner-Wells tongs, incremental weights about 5 lb per level up to about 10 lb/level plus 10 lb for the head, serial neurological examination and radiographs) **is safe without pre-reduction MRI**. **Obtunded/uncooperative patient or failed closed reduction: MRI first** to exclude a **disc herniation** behind the vertebral body (could be pushed into the cord), then open reduction (anterior discectomy first if a disc is present). Then **stabilisation** (anterior cervical discectomy and fusion, or posterior lateral mass fixation).
- **Burst fractures (vertical compression):** retropulsion; anterior corpectomy and fusion if cord compression.
- **Teardrop fracture (flexion-compression): anteroinferior vertebral body fragment** with posterior displacement of the remaining body into the canal and **posterior ligament disruption**: **very unstable**, often **anterior cord syndrome**. Distinguish from the **extension teardrop** (small anteroinferior avulsion of C2 by the ALL in extension: usually stable).

- **Clay-shoveler's fracture:** spinous process avulsion (C7, C6, T1); stable.
- **Ankylosing spondylitis / DISH: extension injuries through the fused spine** ("**chalk-stick**" fractures) are **highly unstable**, frequently missed, often with **epidural haematoma** and late neurological deterioration. **CT/MRI the whole spine; never force into extension** (immobilise in the pre-injury kyphotic position); usually **long posterior fixation** (three levels above and below).
- **Central cord syndrome** (hyperextension in an elderly spondylotic spine, no fracture): see below.
- **Vertebral artery injury:** with **foramen transversarium fractures** and facet dislocations: **CT angiography**; antiplatelet/anticoagulant therapy if no contraindication.

6. Spinal cord injury (SCI)

Assessment

- **ASIA (ISNCSCI) examination: 10 key muscles** (C5 elbow flexors, C6 wrist extensors, C7 elbow extensors, C8 finger flexors (middle finger DIP), T1 little finger abductors; L2 hip flexors, L3 knee extensors, L4 ankle dorsiflexors, L5 long toe extensors, S1 ankle plantarflexors) and **28 sensory dermatomes** (light touch, pin prick), plus **deep anal pressure and voluntary anal contraction (sacral sparing)**.
- **ASIA Impairment Scale:**

Grade	Definition
A	**Complete**: no motor or sensory function in S4-S5
B	Sensory incomplete: sensory but no motor function below the level, including S4-S5
C	Motor incomplete: motor function below the level, **more than half** of key muscles below the level **grade under 3**

Grade	Definition
D	Motor incomplete: **at least half** of key muscles below the level **grade 3 or more**
E	Normal

- **Spinal shock:** temporary loss of all cord function and reflexes below the injury (flaccid paralysis, areflexia), **ends with the return of the bulbocavernosus reflex** (usually within **24-48 hours**). **Completeness can only be judged after spinal shock resolves**: no sacral sparing after the bulbocavernosus reflex returns = complete injury (very poor prognosis for recovery).
- **Neurogenic shock:** loss of sympathetic tone (injuries above **T6**): **hypotension with bradycardia**, warm peripheries. Treat with fluids then **vasopressors** (noradrenaline/phenylephrine), atropine for bradycardia. **Haemorrhagic shock** shows tachycardia and must be excluded first.
- **Maintain mean arterial pressure 85-90 mmHg for about 7 days** (AANS/CNS guideline).
- **Methylprednisolone: not routinely recommended** (NASCIS II/III results not reproducible; complications: infection, GI bleeding). AO Spine 2017 guideline: may be considered within 8 hours (24-hour infusion) as an option; never after 8 hours.
- **Early decompression (within 24 hours)** improves neurological recovery (**STASCIS** study, cervical SCI; AO Spine guideline recommends surgery within 24 hours).

Incomplete cord syndromes

Syndrome	Mechanism	Features	Prognosis
Central cord	**Hyperextension in an elderly spondylotic spine** (often no fracture)	**Upper limbs weaker than lower** (hands worst), sacral sparing, variable sensory loss	**Good**; bowel/bladder and walking usually recover (hands least); **commonest incomplete syndrome**
Anterior cord	Flexion, anterior compression, **anterior spinal artery** injury	**Motor loss and loss of pain/temperature**; **preserved proprioception and vibration (dorsal columns)**	**Worst** prognosis of the incomplete syndromes
Brown-Sequard	Penetrating injury, hemisection	**Ipsilateral motor loss and proprioception loss; contralateral pain and temperature loss** (one to two levels below)	**Best prognosis** (about 90% walk)
Posterior cord	Rare (hyperextension)	Loss of proprioception and vibration only	Good
Conus medullaris	T12-L1 injury	Bowel/bladder/sexual dysfunction, saddle anaesthesia, mixed UMN/LMN; absent bulbocavernosus reflex may persist	Variable
Cauda equina	Below L1	**LMN** pattern: asymmetric lower limb weakness, areflexia, saddle anaesthesia, sphincter dysfunction	Better than conus (peripheral nerves)

Functional levels (complete injuries)

- **C3 and above:** ventilator-dependent (phrenic C3-5).
- **C5:** elbow flexion; power wheelchair (hand control), feeds with adapted devices.
- **C6: wrist extension (tenodesis grip)**; manual wheelchair with adaptations, independent transfers possible.
- **C7: elbow extension (triceps)**: independent transfers and manual wheelchair; **key level for independent living**.
- **C8-T1:** hand function; full wheelchair independence.
- **Upper limb tendon transfers** in tetraplegia: **posterior deltoid or biceps to triceps** (elbow extension), **brachioradialis to ECRB** (wrist extension), **brachioradialis to FPL** (key pinch), nerve transfers.
- **Autonomic dysreflexia** (injuries above T6): noxious stimulus below the injury (**full bladder, faecal impaction**, pressure sores) causes **severe hypertension with bradycardia, headache, flushing/sweating above the lesion**: sit the patient up, remove the stimulus (catheterise), antihypertensives.

7. Paediatric cervical spine

- **Under about 8 years: C1-C3 injuries predominate** (large head, horizontal facets, ligamentous laxity, fulcrum at C2-3); **pseudosubluxation of C2 on C3 (up to about 4 mm)** is normal (**Swischuk line**: posterior arch line C1 to C3 should pass within 2 mm of the C2 spinolaminar line); **ADI up to 5 mm**; **SCIWORA**.
- Use a **backboard with an occipital cut-out or shoulder roll** to avoid neck flexion.
- **Atlantoaxial rotatory subluxation (Fielding-Hawkins types I-IV)**; **Grisel syndrome** (after pharyngeal infection or surgery); **cock-robin** torticollis; dynamic CT. Treat early with collar/traction, halo or fusion for chronic cases.

Exam pearl: **Rule of Spence (lateral mass overhang over about 7 mm) = TAL rupture; Dickman I (midsubstance) needs fusion. Odontoid type II = commonest and highest non-union; IIB = anterior screw, IIC = posterior C1-C2 fusion. Hangman IIA: no traction.**

Exam pearl: **Awake patient with facet dislocation: closed reduction without MRI is safe; obtunded or failed: MRI first. SLIC over 4 = surgery.** Ankylosing spondylitis fractures are highly unstable: CT/MRI whole spine.

Exam pearl: **Central cord (elderly hyperextension, arms worse than legs) is commonest; Brown-Sequard best prognosis; anterior cord worst.** Spinal shock ends with return of the **bulbocavernosus reflex**; neurogenic shock = hypotension + bradycardia (above T6). **C7 = key level for independence.**

Written by Claude for Ortho preSSaleel · updated 2026-11-15. Check doubtful points in a standard textbook.